

Research Paper

A qualitative study of how and why gym-based resistance training may benefit women's mental health and wellbeing

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ABSTRACT

Purpose: This study used photovoice methodology to explore women's lived experiences of gym-based resistance training, and the potential mechanisms by which resistance training may promote wellbeing.

Methods: Ten women aged between 18 and 27 years from various locations across England each generated a total of three photographs illustrating 1) *what they like* and 2) *what they dislike about gym-based resistance training*, and 3) *what gym-based resistance training means to them*, and completed a subsequent photo-elicitation telephone interview via Zoom video conferencing software. Data were analysed through an iterative deductive and inductive process, firstly using Ryff's psychological wellbeing model (Ryff, 1989) as a thematic framework, and then inductively to further explore the data.

Results: Five themes linking gym-based resistance training and mental health and wellbeing were identified: (1) self-acceptance, (2) personal growth, (3) flow state, (4) social affiliation and (5) autonomy. Through engaging in resistance training participants exercise self-direction (i.e., autonomy) and have the opportunity to develop positive relations with like-minded others. The activity itself can facilitate the achievement of the "flow state" of mind. The setting, pursuit and achievement of meaningful "intrinsic goals" can result in feelings of personal growth and realisation a more positive self-regard.

Conclusion: Gym-based resistance training provides various paths to wellbeing. Greater emphasis of these wellbeing benefits in public health messaging may encourage more women to engage in gym-based resistance training.

1. Introduction

Obesity and mental health disorders are worldwide problems with high costs to individuals and society (Goodwin et al., 2020; NCD Risk Factor Collaboration, 2016; Patel et al., 2018). Regular physical activity can prevent obesity and improve mental health (Lee et al., 2012; World Health Organization, 2022). Adults around the world are recommended to participate in muscle-strengthening activities on at least 2 days per week to benefit their health (Bull et al., 2020). Despite the extensive evidence supporting the physical and mental health benefits of participating in muscle-strengthening activities (referred to as resistance training herein; Ashton et al., 2018; Gordon et al., 2017; B.R., 2018; Maestroni et al., 2020; Momma et al., 2022), less than 20% of adults report meeting the muscle-strengthening activity guideline (i.e., ≥ 2 times/week; Bennie et al., 2020).

Resistance training encompasses a range of voluntary activities

including lifting one's body weight (i.e., callisthenics such as push-ups or pull-ups) as well as the use of hand-held weights (i.e., free weights, machine weights, or resistance bands). Such activities can be performed in a range of settings including the home, outdoors, gyms and leisure facilities. On average, women report poorer mental health than men (Boyd et al., 2015; Kuehner, 2017), are less physically active than men (Guthold et al., 2018; Mielke et al., 2018), participate in less resistance training than men (Bennie et al., 2016a; Strain et al., 2016), and are typically underrepresented amongst gym users that do participate in resistance training (Brace-Govan, 2004). Given the mental health benefits of regular resistance training (Gordon et al., 2017; B.R., 2018), promotion strategies to encourage and support women to participate in resistance training may result in substantial public health benefits and contribute to attenuating gender differences in mental health.

Much of the research in this area has focused on the efficacy of resistance training for the treatment of mental health disorders

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including anxiety and depression (Gordon et al., 2017; B.R., 2018; Heissel et al., 2023). Other research has examined barriers to engaging in resistance training. For example, there is consensus amongst some quantitative studies that insufficient time, lack of knowledge and muscle soreness are barriers to resistance training amongst college age women (Hurley et al., 2018; Peters et al., 2019). Other quantitative studies have revealed that resistance training participation motives are largely characterised by extrinsic goals (e.g., appearance; improving physical abilities and performance; Dominski et al., 2020; Marin et al., 2018). However, a key limitation to these and previous quantitative studies (Milton et al., 2018; Sandercock et al., 2022; Strain et al., 2016) is that they offered limited contextual understanding as to women's personal experiences of resistance training.

Qualitative methods offer the opportunity to explore personal experiences of engaging with gym-based resistance training and the meaning women ascribe to it. Of the small number of qualitative studies in this area, most have explored barriers to resistance training engagement, and revealed gendered body ideals, the masculine nature of 'lifting weights', and the gendering of spaces to be key factors (Coen, Rosenberg & Davidson, 2018; Turnock, 2021). A Canadian study for example used a mixture of qualitative methods including drawings, journaling and interviews with men and women to explore the gendered nature of gym-based exercise activities and practices (Coen, Rosenberg & Davidson, 2018). Other qualitative research has focused more on the positive aspects of resistance training by exploring the utility of transformational photography amongst women weightlifters in recovery from eating disorders (Hockin-Boyers, Pope & Jamie, 2021a, 2021b; Hockin-Boyers & Warin 2021). While this evidence is noteworthy, there remains an absence of evidence relating to the positive affective responses and social experiences women ascribe to gym-based resistance training and how these positive experiences can benefit their mental health and wellbeing.

Self-determination theory posits that psychological wellbeing (referred to as wellbeing herein) is universally dependant upon the satisfaction of three basic psychological needs including autonomy, competence, and relatedness (Ryan & Deci, 2000; R.M. 2017; Martela & Ryan, 2021; Martela & Sheldon, 2019). Ryff (1989) proposed a broader set of six core elements of wellbeing including self-acceptance, positive relations with others, autonomy, environmental mastery, purpose in life, and personal growth. The conceptual model's support of multiple theoretical perspectives and its widespread scientific impact make it a suitable framework to explore the potential mechanisms underlying the effects of resistance training on women's mental health and wellbeing (Ryff, 2014; C.D. 2018; Ryff & Singer, 1996). Some qualitative research has explored how the social aspect of CrossFit participation, including the encouragement and support received from others is a potential pathway to improved mental health and wellbeing (Lautner et al. 2021; Vasudevan & Ford 2022). However, there is a real need for a more holistic understanding of women's positive experiences of resistance training including the attainment of wider wellbeing elements and intrinsic goals (e.g., personal growth and self-acceptance), as well as potential simultaneous negative psychological experiences.

Photovoice is a visual participatory research methodology that encourages participants to visually document, reflect upon, and communicate issues of meaning and concern with the view of stimulating social change (Wang & Burris, 1997). Due to the traditional "deficits" framework focus in health research and the overwhelming emphasis on exercise as a tool to alleviate distress and mental health disorders, there is comparatively little research — particularly participatory research (i.e., using photovoice) — that applies a positive lens, and explores how resistance-training in particular enables women to keep well and thrive (VanderWeele, 2017). Therefore, we aim to explore this gap by drawing upon positive psychology and photovoice methodology to capture a richer understanding of women's' positive lived experiences of resistance training, and the potential mechanisms by which resistance training may promote wellbeing. The central question underpinning the

study is: how and why gym-based resistance training may affect women's wellbeing?

2. Materials and methods

2.1. Participants and recruitment

A purposive sample of ten participants aged between 18 and 27 years (mean = 21.1) were recruited from various locations across England (see Table 1 for further information). The majority of the participants were from Liverpool (5 participants). Eighty percent of the participants were students. Three participants were unemployed, five were employed part-time, and the remaining two were employed full-time.

To be eligible to take part in the study participants had to be female, aged at least 18 years, fluent in English, and have experience of regular gym-based resistance training. Participants were also required to confirm that they had access to a mobile device to take photographs, and had the availability and capacity to complete an interview via Zoom video conferencing software at a later date. In the first instance, study recruitment posters detailing the nature of the study, eligibility criteria and the first author's email address were placed in various gym settings in the city of Liverpool. Because participant recruitment was initially low study information was subsequently circulated on social media platforms (i.e., Instagram and Facebook) to increase recruitment outreach. Interested persons emailed the first author and were subsequently provided with a participant information sheet and a consent form to complete and return via email. All participants were entered into a prize draw with the opportunity to win a £40 Amazon e-voucher. The study received ethics approval from the University of Liverpool (reference number: 10310) and data collection occurred between November 2021 and January 2022.

2.2. Data collection

This qualitative study adopted a phenomenological approach. We used Photovoice methodology (Wang & Burgess 1997) alongside individual interviews conducted via Zoom video conferencing software to encourage participants to think critically about their gym-based resistance training experiences. Photovoice is a participatory visual research methodology that utilises photography and narratives to explore social issues. By adopting this participatory visual approach, the participants were provided with a high degree of control over their expression, allowing them time to reflect upon and articulate what is most important to them. Moreover, the photographs provide a rich visual illustration representing the participant's lived experiences (Wang et al. 1998; Budig et al. 2018).

After providing written informed consent, each participant used their own digital camera to take a total of three photographs illustrating 1) 'what they like and 2) what they dislike about gym-based resistance training', and 3) 'what gym-based resistance training means to them'. In order to reduce the researchers' influence over the participant's choice of photographs, the participants did not receive specific guidance or

Table 1
Participant characteristics.

Participant ID	Age in years	Home city	Employment status	Student
F1	19	London	Part time	Yes
F2	20	Liverpool	Unemployed	Yes
F3	20	Liverpool	Part time	Yes
F4	22	Cambridge	Unemployed	Yes
F5	18	Liverpool	Unemployed	Yes
F6	20	Liverpool	Part time	Yes
F7	21	Leeds	Part time	Yes
F8	24	Southampton	Full time	No
F9	27	Norwich	Full time	No
F10	20	Warrington	Part time	Yes

examples of potential photographs that they could take. However, the ethical implications of individuals appearing in photographs was discussed with participants. Participants were instructed to not take photographs of any other person and were informed that any photograph that captured their own face would be anonymised in future publications. Once the participant had emailed their three photographs to the first author an interview was scheduled.

Interviews incorporating a photo-elicitation reflective activity were conducted by the first author. A semi-structured interview guide was developed and used across all interviews to ensure consistency. The second author has experience conducting qualitative research on similar topics to that of the present study (Bhandel & Noonan 2022; Cox, Fairclough & Noonan 2021; Noonan et al. 2016; R.J. 2017). The structure and content of the interview guide were informed by these aforementioned studies. Prior to data collection the second author piloted the questions and activities with adults of similar age to the present study. Feedback was subsequently received on interview structure and question appropriateness, and a couple of the questions in the original interview guide were rephrased.

Interviews lasted roughly 30 min (mean = 29.8). The first author reiterated to the participants at the beginning of the interview that they were free to withdraw their participation without reason at any point throughout the interview, and that their appearance could remain anonymous if they did not feel comfortable showing their face on camera. All participants opted to show their face on camera. The opening section of the interview addressed demographic questions and comprised an icebreaker activity relating to recent physical activity experiences for the purpose of building rapport between researcher and participant.

The photo-elicitation reflective activity was then completed which provided the participants with an alternative way of sharing their lived experiences of gym-based resistance training. The interactive activity centred on 'what the participants like and what they dislike about gym-based resistance training', and 'what gym-based resistance training means to them'. Zoom video conferencing software has screen-sharing capabilities both for the researcher and participant, and this function was used to display photographs during the photo-elicitation aspect of the interview. For each of the photographs the first author used prompts based on the commonly used SHOWeD technique (Catalani & Minkler 2010; Wang 1999) to encourage participants to articulate their own meaning embedded within the photograph. The SHOWeD technique prompts included: What do you See here? What is really Happening here? How does this relate to Our lives? Why does this situation, concern, or strength exist? What can we Do about it? (Wang, 1999). The final section of the interview comprised additional open-ended questioning aligned to how and why gym-based resistance training may affect women's mental health and wellbeing. For example, when you resistance train at the gym how does it make you feel at the time, and how do you feel after you've resistance trained at the gym? On completion of the interview participants were provided with a debrief form. The debrief form contained the contact details of the authors and signposting to GP services should they be concerned about their health and wellbeing after participating in the study. Interviews were audio-recorded via Zoom, and were transcribed verbatim for further analysis and anonymised using pseudonyms (i.e., female (F) and ID number).

2.3. Data analysis

Two separate but related data sources were generated and analysed – visual data from the photovoice activity and verbatim data from the interview and photo-elicitation narratives. The two data sources were combined and a mixed analysis approach was taken for complimentary purposes. A thematic analysis approach was taken to analyse the data. The procedures set out by Braun and Clarke (2006) were followed. After listening to the interview recordings and reviewing the transcripts, the first author adopted a deductive and inductive approach to generate a

series of codes which were subsequently collated and organised into themes aligned to the study aims. In the first instance, Ryff's psychological wellbeing model (Ryff, 1989) was used as a thematic framework, and then data were further analysed inductively. The process was repeated and an iterative approach was used to ensure the themes told an accurate, coherent and compelling story of the data (Braun, Clarke & Weate 2016). A similar analysis approach was taken for the photo-elicitation data. The photo-elicitation narratives were classified as a written 'report' and appended to each individual photograph. The narrative reports and photographs were then used in combination to categorise the visual data in relation to specific themes. The narrative complementing participants' photographs both added context and verified the content in photographs. By triangulating the two data sources researcher biases were limited and credibility was enhanced. The transcripts and photographs were independently reviewed by the second author, and were subsequently cross-examined against the data in reverse, from the themes to the codes and data sheets. This process ensured accuracy and facilitated alternative interpretations of the data. The authors then created thematic maps and completed the written report, which were discussed, reflected upon, refined and approved (Braun & Clarke, 2006).

3. Results

Five themes linking gym-based resistance training and mental health and wellbeing were identified (Refer to Fig. 1). Themes covered self-acceptance, personal growth, flow state, social affiliation and autonomy.

3.1. Self-acceptance

Participants noted an improved sense of self (i.e., self-esteem and confidence) from engaging in resistance training and reported that by developing a more positive attitude to themselves they were motivated to push themselves further and adhere to their training goals. The participants explained how resistance training provides them with opportunities for challenge and a platform to realise one's potential and to demonstrate to themselves that they are 'capable of more than they believe' which gives them a sense of empowerment (Refer to Fig. 2).

I definitely feel more confident and prouder of myself. It motivates me to keep doing it everyday because I can look back and see what I've done on certain days and then I think to myself that maybe I can beat it ... With looking at yourself in the mirror you won't see changes and stuff so I feel happy comparing myself in terms of progress in this way so I can see that obviously if my recovery time is getting less then I know that I'm getting fitter (F5).

One participant in particular, spoke vividly about the empowerment she gained from resistance training and how she no longer requires medication to manage her mental health disorder.

I was quite depressed and anxious and didn't realise [at the time] the benefits of physical exercise for depression. I got off medication because I didn't need it anymore after gymming. It made me so happy to feel so strong and empowered by myself. I became content with my life and the only thing that had changed was me going to the gym (F4).

Other participants reported improved self-esteem and confidence from experiencing aesthetic changes to their body which gave them extra reassurance in the way they look and feel about themselves (Refer to Fig. 3).

3.2. Personal growth

Participants expressed a desire to continually progress the intensity of their resistance training workouts in order to achieve personal growth and seek mastery (Refer to Fig. 4 and Fig. 5). The feeling of getting better and better at something that matters was cited as a key source of

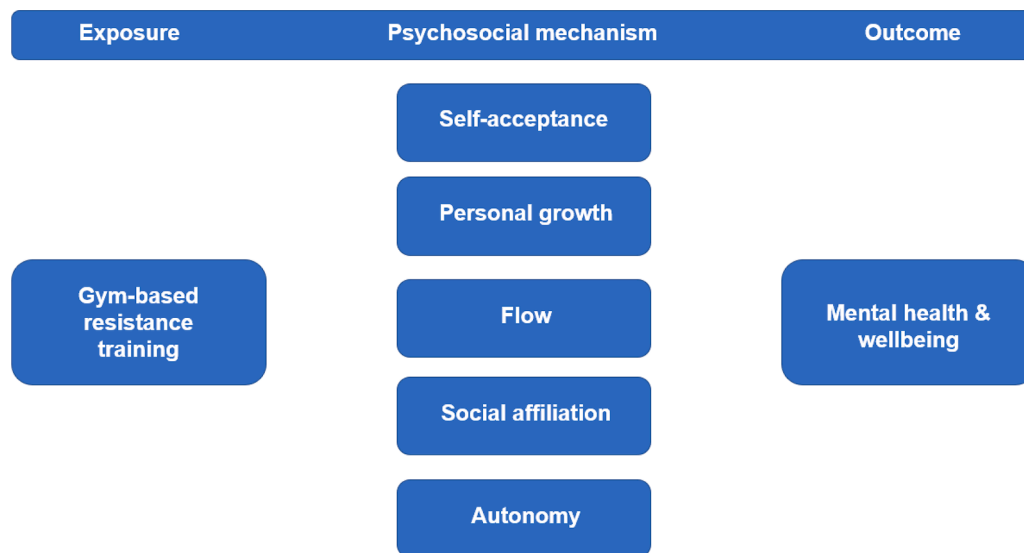


Fig. 1. Thematic map illustrating themes linking gym-based resistance training and mental health and wellbeing.



I had just hit my personal best on squat. It was a moment where I realised, I actually really enjoy this. The feeling I got from hitting the PB was a feeling I hadn't ever had. I just felt so strong and confident, and I just thought I'm going to take a photo because I'm so proud of myself. It's just the feeling of knowing I can lift heavy stuff and still enjoy it... The confidence I get from it and the empowerment is like nothing else (F2).

Fig. 2. Photograph illustrating feelings of self-acceptance.

enjoyment and wellbeing for the participants. Participants highlighted that the intensity of their workout is an important feature of an enjoyable workout experience. They recalled how the intensity of their resistance training workout is used as an indicator of their training progress, and gives them a sense of how much they have improved, and how far they are able to push their body and achieve (i.e., physical abilities).

I like to see myself growing in strength ... it makes me feel like I'm progressing, and it gives me a challenge to see how heavy I can lift and tracking the weights increase motivates me to keep going because I can see that I'm getting somewhere. The fact that I know I'm getting stronger is the best motivation. The thing I'm most proud of is that these weights are for my upper body which is an area I have been wanting to grow. My strength in my upper body has increased a lot and improvements in my form have helped with my progress, so this pushes me to keep training (F7).



Changing the way my body looks and feeling in good shape plays a huge part in my confidence ... I feel a lot more confident when I look and feel good, on the inside and outside (F9).

Fig. 3. Photograph illustrating feelings of self-acceptance.

3.3. Flow state

Participants explained how resistance training at the gym was a form of *escapism*. The intense concentration required when focusing on specific training goals and movements not only provides the participants with the 'headspace' they desire, but also enables the achievement of an ordered state of mind which improves their psychological functioning (Refer to Fig. 6).

It gives you an hour and a half in a day just clear your head and focus on yourself, like an escape. With weight training you have to be so focused



I like how strong I look in it ... It shows my progress with weightlifting. I look strong, and my form is pretty good considering where I was a year ago. I love the progress with weightlifting ... I feel proud of myself to keep training and pushing myself and giving myself personal goals. I love just doing it for myself and I feel good in myself (F1).

Fig. 4. Photograph illustrating feelings of personal growth.



I love this photo because it shows me stepping out of my comfort zone and attempting hip thrusts which I previously have shied away from ... I still have far to go but that's what's so exciting about the gym, there is always progress to be made. I took this photo to check form and to hype myself up ... which is a huge motivating factor sometimes. This photo tells us that my weight training is progressive. There is plenty of room for improvement and growth, but I am really enjoying the journey (F10).

Fig. 5. Photograph illustrating feelings of personal growth.

and so careful and if you're thinking about something else then it can be dangerous (F5).

Going on a walk is nice and clears your head but going to the gym gives a good sense of self achievement and mentally benefits me a lot more than a



You know how a rainbow comes after a rainy day. Before I started weightlifting, I was just doing cardio because I was trying to lose weight but I ended up doing it in the most unhealthy way possible I started doing weight training and realised I really like this. So, the rainbow symbolises this basically, and how the weightlifting was the 'pot of gold' for me to keep going, it's my escape (F2).

Fig. 6. Photograph illustrating feelings of flow.

walk ever would ... When you go you're there for a reason and you can dissociate from the outside world. There are no distractions or things that take away from you getting it done while there (F3).

Participants spoke about their experience of using resistance training as a coping mechanism for managing everyday stressors, 'clearing their worries' and overcoming life challenges.

I see it as my escape and a way to shut off from the stress and pressure of outside world ... It gives me a break from uni and nursing ... There's so much pressure to do work, go out and drink ... [Training] gives you something to focus on just for you and time to be in your own head. It allows me to be healthy physically and mentally. I'd say mentally it eases stress more than explicitly making me happy (F7).

3.4. Social affiliation

Gym-based resistance training provided participants with opportunities for social affiliation. A majority of participants reported feeling more motivated to resistance train around friends, the importance of having friends for support and encouragement, and the enjoyment that results from connecting with and training with people sharing similar interests and shared goals. The competitive nature of some forms of resistance training like CrossFit was attractive to some participants as it enabled them to push themselves further and achieve a higher level of performance than if they were to train alone. Other participants reported that the community nature of the gym setting was vital to learning more about resistance training, and enables them to acquire the information and assistance they need to achieve their resistance training goals.

Gym-based weight training can be incredibly inclusive and sociable and provide real opportunities to meet new like-minded people. I tend to prefer, and spend more time training with other people ... I like to have people around to motivate me, help with technique and it provides a social aspect to weight training (F8).

However, for some participants developing social networks at the gym, training with others (i.e., co-participation), receiving social support, and using resistance training as an affective and emotional resource was not considered important to their training experience and

wellbeing. These particular participants highlighted a preference for training alone as training with others limited their autonomy and freedom to train at times that best suited their lifestyle schedule, and enabled them to experience ‘quiet time’.

Sometimes I just want to go to the gym to clear my head. I don't want to have to chat ... if I've had a bad day (F5).

3.5. Autonomy

While the participants exhibited autonomy by way of regulating their behaviour, resisting social pressures surrounding resistance training, and assessing themselves against personal standards rather than the expectations of others, participants noted concerns regarding the evaluations of other gym users. Concerns regarding the evaluations of other gym users cut across two related areas including gender stereotypes and stigma surrounding resistance training and the ‘male gaze’. These concerns were primarily expressed based on the participants initial engagement with gym-based resistance training ahead of them developing confidence, a stronger sense of self, and an internal locus of evaluation.

Gender norms and stereotypes ... and negative comments don't tend to bother me because I feel good in myself and I know the gym is a good thing for me, physically and mentally. But for others it's important to consider the impact that others judgement would have over their approach to weight training (F3).

Groups of men, the staring and making comments ... doesn't really affect my training, but it does make me feel uncomfortable (F4).

[Initially] I recognised weightlifting to be a predominantly male [thing] and I felt worried stepping out of my comfort zone and entering a space I felt wasn't for me to use (F10).

4. Discussion

This study used photovoice methodology to explore women's lived experiences of gym-based resistance training, and the potential mechanisms by which resistance training may promote wellbeing. In doing so we identified five themes and offer contextual insight into the mental health and wellbeing benefits women ascribe to gym-based resistance training.

A recurring theme throughout the data was the wellbeing benefits achieved through gym-based resistance training. Participants explained how following the uptake and engagement in gym-based resistance training their mental health and wellbeing had improved. In some instances, participants recalled being able to transition off prescription medication for their mental health. Other participants described their gym-based resistance training as a mental ‘escape’, a tool for frustration and stress relief. This finding corresponds with the work of Csikszentmihalyi (1990) on the “flow” state: the psychological condition of being cognitively and emotionally absorbed in activities. In the context of the present study, participants experience intense cognitive focus and alertness when resistance training which leaves limited attention left over to worry about negative life events and problems. In addition to the exclusion of negative thoughts, the concentration of the flow experience – together with the pursuit of clear training goals and immediate feedback enables participants to achieve an ordered state of mind that is highly enjoyable. Within self-determination theory, autonomous forms of motivation such as intrinsic motivation exist when behaviour is perceived as enjoyable (Ryan & Deci, 2000; R.M. 2017). Although flow states have been investigated in some exercise and sporting activities including running (Jackman et al. 2019; Stamatelopoulou et al. 2018; Stenseng et al. 2023) additional research is warranted in the field of resistance training, specifically across resistance training modalities (e.g., strength training, CrossFit, and Powerlifting).

It was evident from the women's narratives that resistance training functions as an activity that enables them to achieve growth, experience accomplishment and competence, and develop a positive self-regard. Women expressed a sense of enjoyment as well as feelings of increased confidence and improved psychological function (i.e., wellbeing) from gaining strength, becoming “fitter” and experiencing aesthetic changes to their body. These positive self-perceptions and expectancy of positive outcomes further served as motivators for the women to ‘lift heavier weights’ and pursue mastery. For the women in our study, higher intensity resistance training was particularly appealing as it was used as a measure of personal growth, and was perceived to be more health enhancing relative to lower intensity resistance training. These findings align closely with a range of wellbeing and flourishing theories (e.g., Diener et al., 2010; Ryan & Deci, 2000; R.M. 2017; Seligman, 2011) including the work of Ryff (1989). They also complement two recent qualitative studies. The first study illustrated the benefits of resistance training for women's body image (Walters & Hefferon 2020). The second study evidenced the psychological benefits of transformation photographs representing bodily changes (i.e., look, shape or size) for women weightlifters during eating disorder recovery (Hockin-Boyers et al. 2021a).

While qualitative evidence relating to positive psychological functioning is limited, various systematic reviews have evidenced the efficacy of resistance training in the treatment of mental health disorders including anxiety and depression (Gordon et al. 2017; B.R. 2018). Several experimental studies have demonstrated the anxiolytic effects of single bouts of resistance training, with stronger effects observed for resistance training at low to moderate intensities (i.e., 50–70% 1 rep max (RM)) relative to higher intensities (i.e., >70% 1RM; Strickland & Smith 2014). In this study we have further demonstrated the various potential mechanisms linking gym-based resistance training and wellbeing, beyond the neurobiological pathway (i.e., via changes in brain chemistry including changes in cortisol activity; Dishman et al. 2006; De Nys et al. 2022), and the expectancy of improved psychological health (Lindheimer et al. 2019). In addition to achieving a flow state, engaging in resistance training may lead to a more positive self-regard (i.e., self-esteem and self-efficacy) which can serve as a protective factor in the face of stress and adversity. Highlighting these psychological and wellness features of resistance training rather than simply the pursuit of weight reduction and maintenance goals may be a more powerful and resonating message to communicate to women in physical activity promotional strategies. Physical activity participation levels are especially low amongst individuals experiencing mental health disorders (McDowell et al. 2018). As mental health disorders tend to arise during adolescence and persist into adulthood (Lieb, Becker & Altamura 2005), resistance training intervention programmes at this age may be most appropriate when physical activity participation levels are also known to decline (Guthold et al. 2020).

There is also potential for gym-based resistance training to benefit wellbeing through a social pathway through increased social interaction and development of positive relations with others. Humans have an innate desire to bond and be in the company of others (Allen, 2020). Indeed, the social aspect of gym-based resistance training and the ‘encouragement’ and ‘support’ received from other gym users was perceived as a benefit to mental health and wellbeing which aligns closely with wellbeing and flourishing theories (e.g., Diener et al., 2010; Ryan & Deci, 2000; R.M. 2017; Ryff, 1989; Seligman, 2011) and social cognitive theory (Bandura 1986). Gym partners provided social support in the form of co-participation, and their presence created a sense of belonging. Gym partners were also valued for the technique advice and assistance, and feedback they provided. Recent qualitative research has revealed that CrossFit members assign great value to the community support, encouragement, coaching and sense of camaraderie they experience from CrossFit practice (Whiteman-Sanderland, Hawkins & Clayton 2018; Lautner et al. 2021). However, as we revealed in the present study there is likely to be variability in women's preference to

train alone vs training with others. Factors such as the desire for autonomy and control over training regimes will play some role in this decision making and should be considered in future intervention programmes.

Engaging in gym-based resistance training is by nature autonomous and self-regulating. Participants in the present study resisted social norms and gender stereotypes surrounding resistance training, they set personal intrinsic goals and demonstrated an internal locus of evaluation which together are core facets of wellbeing (Ryff, 1989). Our findings relating to the negative wellbeing effects of gender social norms and evaluation from others complements recent qualitative research showing gendered body ideals, 'hyper-masculine' gym cultures, and the masculine nature of 'lifting weights' to be key barriers to resistance training engagement and experience (Coen, Rosenberg & Davidson 2018; Turnock 2021; Vasudevan & Ford 2022). As reported by women in this study, the presence of other women participating in resistance training in the gym challenges gender stereotypes, and not only normalises the behaviour, but likely reduces heightened concerns surrounding social judgement. While digital fitness trends like 'fitspiration' have contributed to a reduction in stigma surrounding resistance training amongst women (Andreasson & Johansson 2014; Raggatt et al. 2018), they have likely added to gender stereotypes surrounding body ideals and aesthetic appearance (Boepple et al. 2016; Tiggemann & Zaccardo 2015). For women that do not resistance train, such factors are likely to further reduce their motivations and self-perceptions and serve as barriers to gym-based resistance training engagement. While efforts to change broader culture and media messaging is difficult, public health communication campaigns, and gym marketing could focus much more on challenging these negative stereotypes, and endorsing the psychosocial opportunities and benefits that resistance training provides.

In this study we did not explore how individual characteristics (e.g., age, education, weight status, fitness level and history), the COVID-19 pandemic and seasonal factors such as the weather, as well as the frequency, intensity, time, type and context of resistance training influence women's resistance training and wellbeing experiences. These factors should be considered in the future research. In particular, further longitudinal research is warranted to better understand whether specific resistance training modalities confer greater perceived and actual physical and psychological health benefits than others. A majority of participants in the present study were students. Given the high level of confidence and competence of the women in this study, coupled with the geographical and financial barriers to gym-based resistance training, and the self-selecting bias this creates, it is unlikely that the perceptions and experiences of the women in our study are reflective of the wider female population. A current research challenge is to understand how resistance training can be promoted amongst subgroups of the population least likely to engage in physical activity, such as low-income populations (Chastin et al. 2020). Concessionary schemes in leisure facilities have proven effective in improving overall physical activity levels and reducing health inequalities (Higginson et al. 2018), and could be an effective public health strategy to promote and support resistance training amongst these population sub-groups. There are however low-cost alternatives to gyms (i.e., home and outdoors) and further research is warranted to understand women's perceptions and experiences of resistance training in these settings.

We used Photovoice methodology to explore women's lived experiences of gym-based resistance training, and the potential mechanisms by which resistance training may promote wellbeing. Here we provide further evidence of how gym-based resistance training can benefit women's mental health and wellbeing. Through engaging in resistance training participants exercise self-direction (i.e., autonomy) and have the opportunity to develop positive relations with like-minded others. The activity itself can facilitate the achievement of the "flow state" of mind. The setting, pursuit and achievement of meaningful "intrinsic goals" can result in feelings of personal growth and realisation a more

positive self-regard. Therefore, gym-based resistance training provides various paths to wellbeing. Greater emphasis of these wellbeing benefits in public health messaging may encourage more women to engage in gym-based resistance training.

Author contributions

Both authors contributed to the article

Data availability statement

Due to the nature of this research, participants of this study did not agree for their personal data to be shared publicly, so supporting data is not available.

CRediT authorship contribution statement

Felicity C. Hall: Conceptualization, Methodology, Data curation, Writing – review & editing. **Robert J. Noonan:** Conceptualization, Methodology, Supervision, Writing – original draft.

Declaration of Competing Interest

The authors report there are no competing interests to declare.

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